



Continuous Assessment for Laboratory / Assignment sessions

Academic Year 2024-25

Name: Gargi Sukhatankar

SAP ID: 60019220125

Course: Mobile Security and Forensics

Course Code: **DJS221CC6012**

Course: Mobile Security and Forensics Laboratory

Lab Code: **DJS221CL6012**

Year: T.Y. B.Tech.

Sem: VI

Department: Computer Science and Engineering (IoT and Cyber Security with Block Chain Technology)

Performance Indicators (Any no. of Indicators) (Maximum 5 marks per indicator)	1	2	3	4	5	6	7	8	9	10	11	Σ	A vg	A 1	A 2	Σ	A vg
Course Outcome	1	2	2	3	3	4	5	6									
1. Knowledge (Factual/Conceptual/Procedural/ Metacognitive)	5	5	5	5	5	5	5	5	5					5	5		
2. Describe (Factual/Conceptual/Procedural/ Metacognitive)	5	5	5	5	5	5	5	5	5					5	5		
3. Demonstration (Factual/Conceptual/Procedural/ Metacognitive)	4	4	4	5	5	4	5	5	5					5	5		
4. Strategy (Analyse & / or Evaluate) (Factual/Conceptual/ Procedural/Metacognitive)	4	4	4	4	4	4	4	4	4					4	4		
5. Interpret/ Develop (Factual/Conceptual/ Procedural/Metacognitive)	-	-	-	-	-	-	-	-	-					-	-		
6. Attitude towards learning (receiving, attending, responding, valuing, organizing, characterization by value)	4	4	4	4	4	4	4	4	4					4	4		
7. Non-verbal communication skills/ Behaviour or Behavioural skills (motor skills, hand-eye coordination, gross body movements, finely coordinated body movements speech behaviours)	-	-	-	-	-	-	-	-	-					-	-		
Total	21	22	22	23	23	22	23	23	23					23	23		
Signature of the faculty member	Q	Q	Q	Q	Q	Q	Q	Q	Q					Q	Q		

Outstanding (5), Excellent (4), Good (3), Fair (2), Needs Improvement (1)

Laboratory marks Σ Avg. = <u>22</u>	Assignment marks <u>23</u> Σ Avg. =	Total Term-work (25) = <u>22</u>
Laboratory Scaled to (15) = <u>13</u>	Assignment Scaled to (10) = <u>9</u>	Sign of the Student: <u>Gargi Sukhatankar</u>

Signature of the Faculty member:

Signature of Head of the Department



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

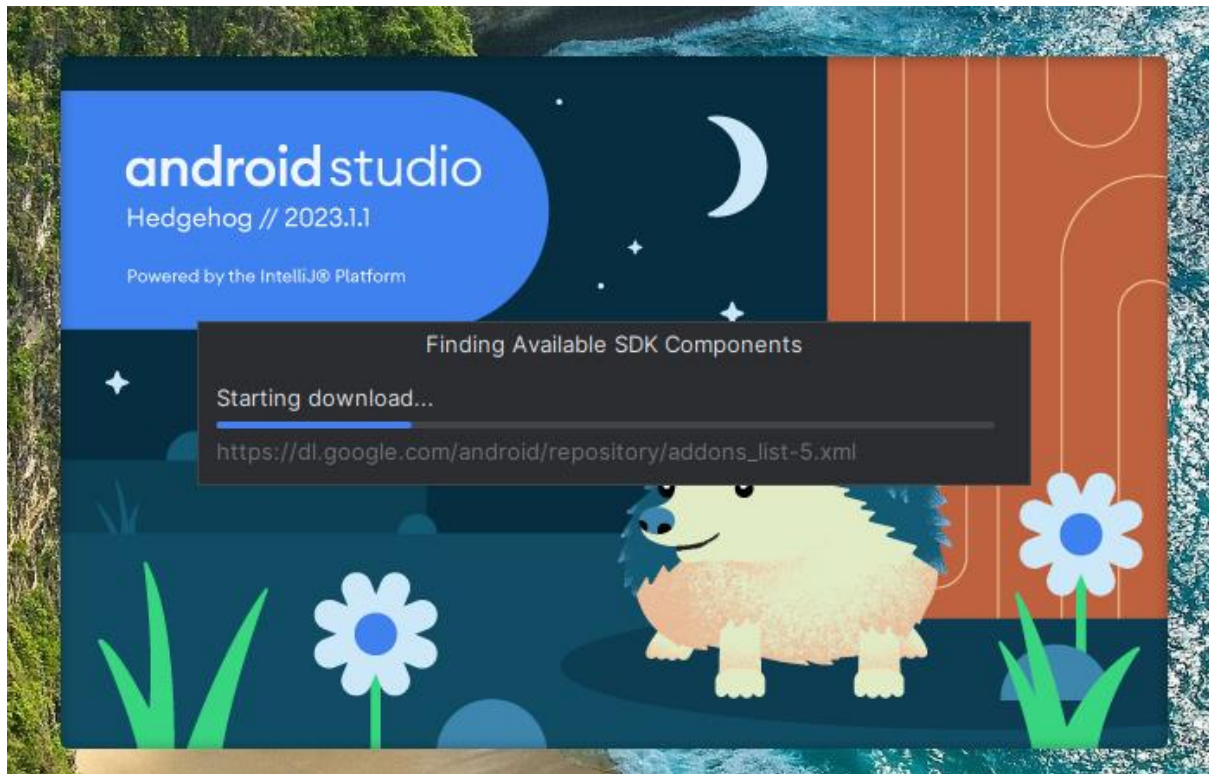
EXPERIMENT 1

Aim: Installation of Android studio with apk installation in VM.

Theory:

Screenshots:

Step 1:



Step 2:



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(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)

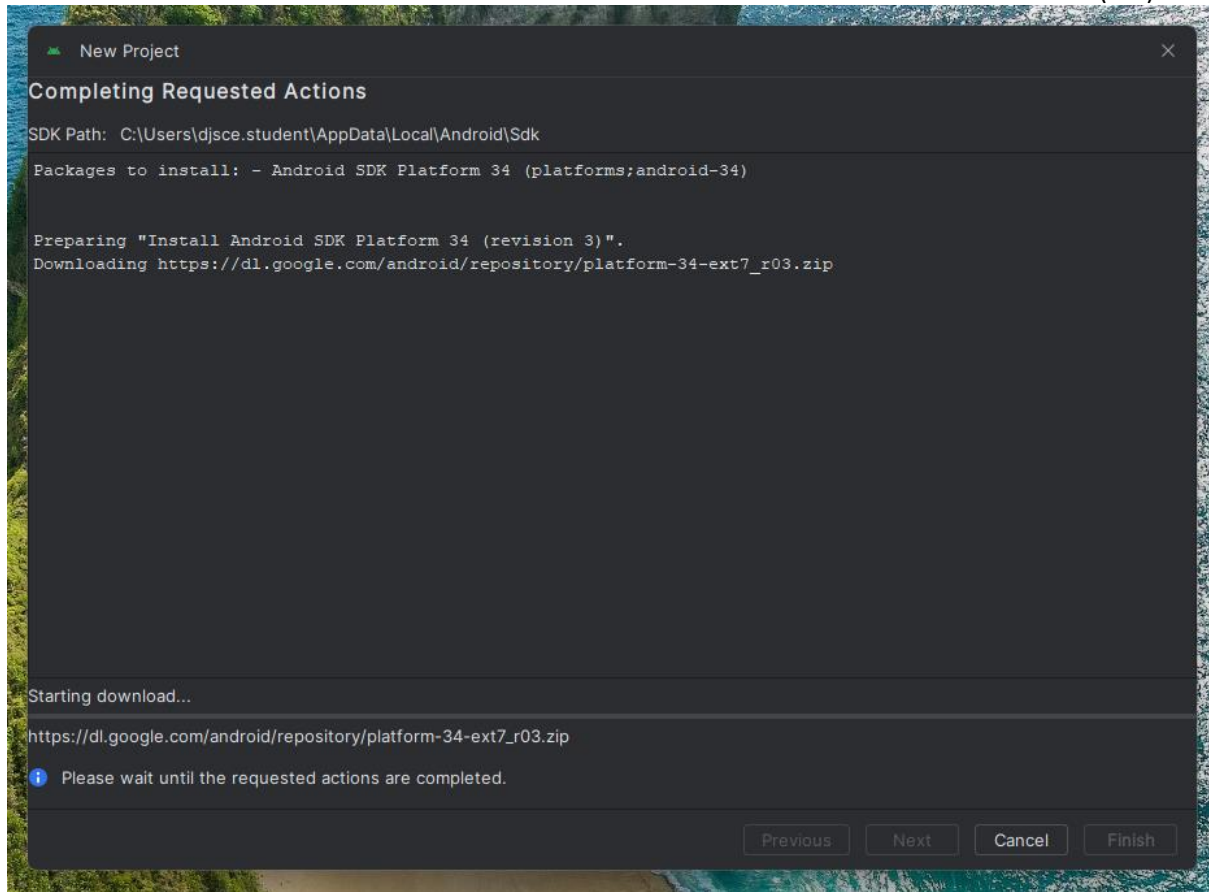


Name: Gargi Sukhatankar

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Roll No: B027

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Step 3:



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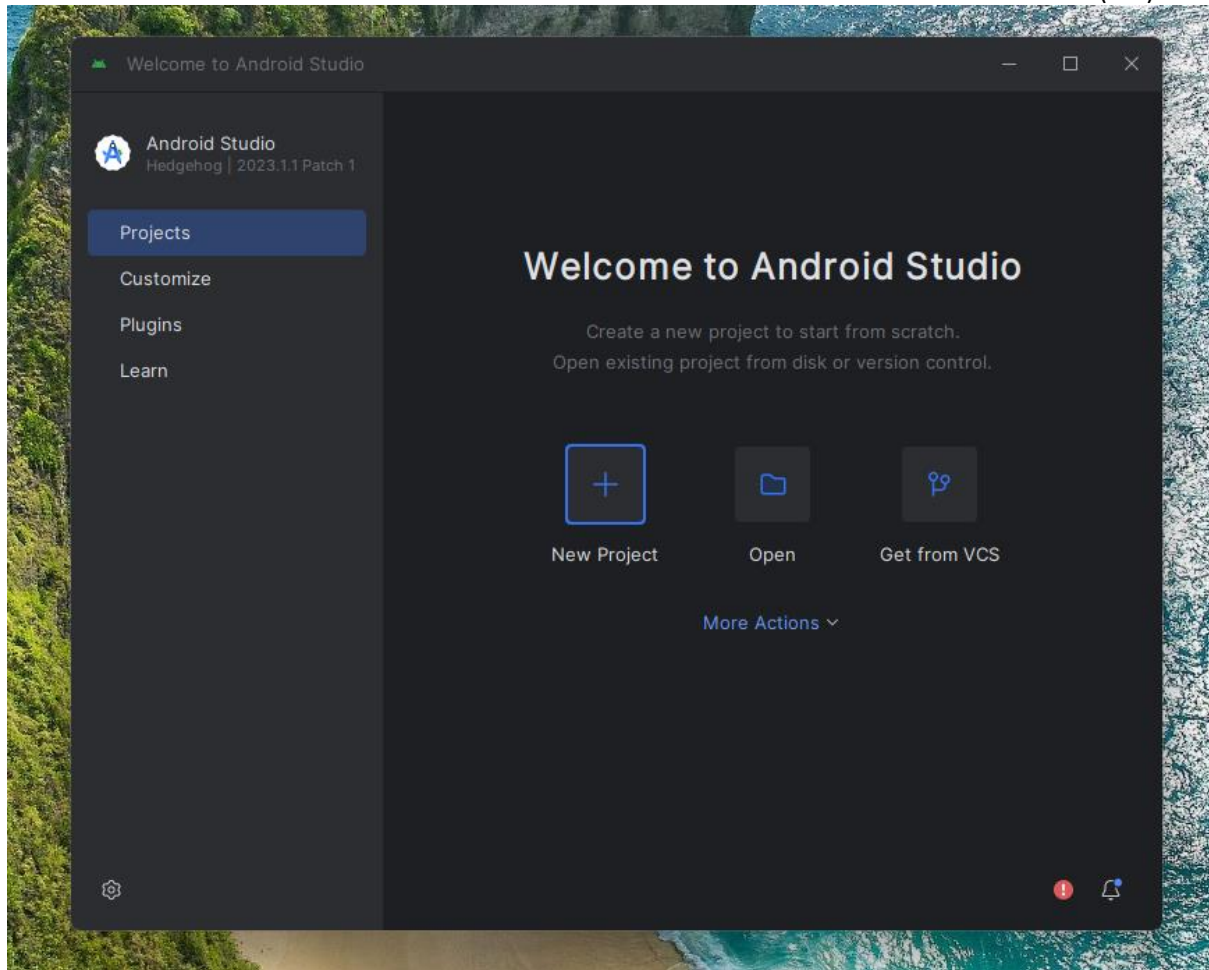


Name: Gargi Sukhatankar

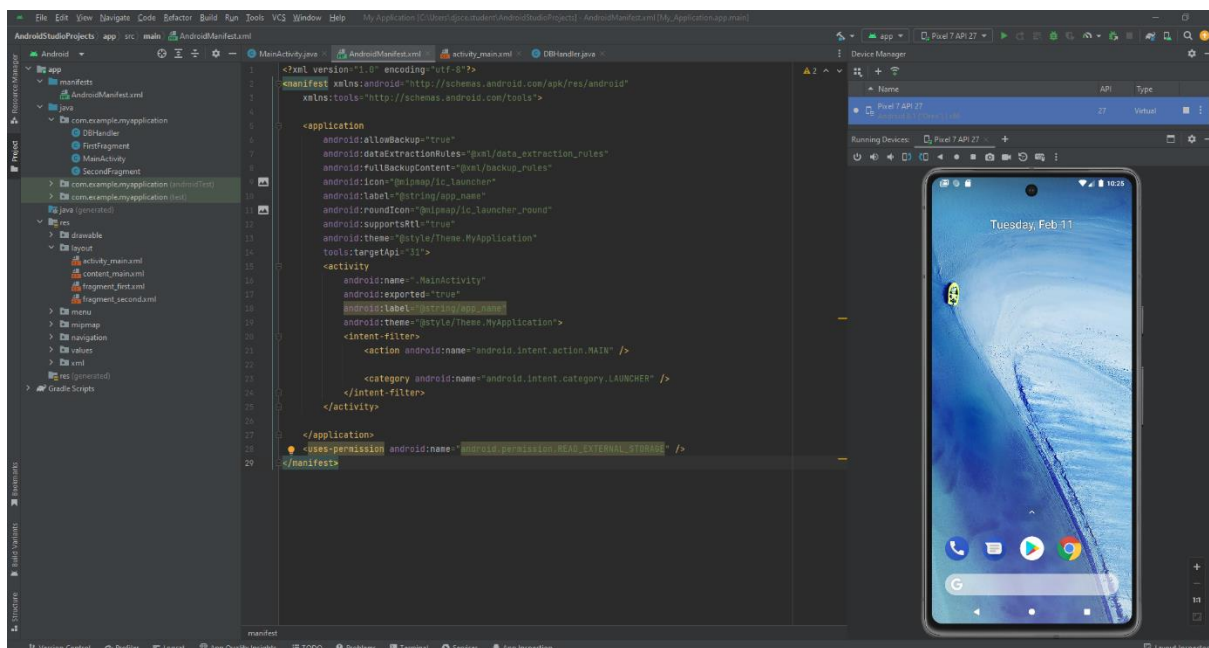
Roll No: B027

SAP ID: 60019220125

Branch: CSE(ICB)



Step 4:





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Roll No: B027

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Branch: CSE(ICB)

Step 5:



Conclusion:

Thus we successfully installed android studio through vmware and downloaded apks on it.

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EXPERIMENT 2

Aim: Installation of Androguard and analysis on apk files.

Screenshots:

Step 1:

```

File Actions Edit View Help

Summary:
Upgrading: 44, Installing: 17, Removing: 2, Not Upgrading: 2083
Download size: 18.9 MB / 52.9 MB
Space needed: 19.9 MB / 61.7 GB available

Get:1 http://kali.kali.org/kali kali-rolling/main amd64 libpython3.12-dev amd64 3.12.8-5deb1 [5,129 kB]
Get:2 http://kali.kali.org/kali kali-rolling/main amd64 python3.12-minimal amd64 3.12.8-5deb1 [2,164 kB]
Get:3 http://mirror.freeweb.org/kali kali-rolling/main amd64 libpython3-dev amd64 3.12.8-1 [18.1 kB]
Get:4 http://kali.kali.org/kali kali-rolling/tftp amd64 2.10-ds-samba-2.1.3dfsg-3 [46.5 kB]
Get:5 http://kali.kali.org/kali kali-rolling/main amd64 python3-talloc amd64 2.10-ds-samba-2.1.3dfsg-3 [46.2 kB]
Get:6 http://kali.kali.org/kali kali-rolling/main amd64 samba amd64 2:4.21.3dfsg-1 [1,378 kB]
Get:7 http://kali.kali.org/kali kali-rolling/main amd64 samba-common-bin amd64 2:4.21.3dfsg-1 [1,123 kB]
Get:8 http://kali.kali.org/kali kali-rolling/main amd64 smbclient amd64 2:4.21.3dfsg-1 [474 kB]
Get:9 http://kali.kali.org/kali kali-rolling/main amd64 python3-idn amd64 2:12.10.0-samba-2.1.3dfsg-1 [71.5 kB]
Get:10 http://kali.kali.org/kali kali-rolling/main amd64 libldb amd64 2:12.10.0-samba-2.1.3dfsg-1 [173 kB]
Get:11 http://kali.kali.org/kali kali-rolling/main amd64 samba-adu-middleware amd64 2:4.21.3dfsg-1 [133 kB]
Get:12 http://kali.kali.org/kali kali-rolling/main amd64 samba-libc amd64 2:4.21.3dfsg-1 [5,961 kB]
Get:13 http://kali.download.com/kali kali-rolling/main amd64 python3-nsslib amd64 5.1.0-0-kali1 [1,736 kB]
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Get:15 http://kali.download.com/kali kali-rolling/main amd64 onboard amd64 1.4.1-9 [150 kB]
Get:16 http://mirror.skx.ru/kali kali-rolling/main amd64 python3-dev amd64 3.12.8-1 [26.1 kB]
Get:17 http://kali.download.com/kali kali-rolling/main amd64 blossom amd64 2.4.3-1 [1,019 kB]
Get:18 http://kali.download.com/kali kali-rolling/main amd64 python3-amf64 3.12.0-1 [27.9 kB]
Get:19 http://kali.kali.org/kali kali-rolling/main amd64 libldb amd64 2.10-ds-samba-2.1.3dfsg-1 [76.4 kB]
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Get:30 http://kali.kali.org/kali kali-rolling/main amd64 python3-arcd amd64 0.3.0-0-kali1103 [7,832 B]
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Get:92 http://mirrors.neusoft.edu.cn/kali kali-rolling/main amd64 libpython3-stdb amd64 3.12.8-1 [1]
Get:93 http://kali.kali
```

```

Downloading androguard-4.1.2-py3-none-any.whl (950 kB)
950.2/950.2 kB 6.4 MB/s eta 0:00:00
Downloading apkinspector-1.3.2-py3-none-any.whl (29 kB)
Downloading dataset-1.6.2-py2.py3-none-any.whl (18 kB)
Downloading frida-16.6.6-cp37-ab13-manylinux2_5_x86_64.whl (29.8 MB)
29.8/29.8 MB 6.2 MB/s eta 0:00:00
Downloading loguru-0.7.3-py3-none-any.whl (61 kB)
61.6/61.6 kB 335.2 kB/s eta 0:00:00
Downloading banal-1.0.6-py2.py3-none-any.whl (6.1 kB)
Building wheels for collected packages: mtf8
  Building wheel for mtf8 (setup.py) ... done
  Created wheel for mtf8: filename=mtf8-1.0.6-cp311-cp311-linux_x86_64.whl size=19248 sha256=a3ab7bb7fc9b253f00d6b116f736ad09f0bc8a410b70e1511cff713683ba2c
  Stored in directory: /home/kali/.cache/pip/wheels/44/bc/d6/647be2901ee64bf5bbc7636ea53f146947a204f61d552771d3
Successfully built mtf8
Installing collected packages: mtf8, banal, loguru, frida, apkInspector, dataset, androguard
WARNING: The script apkInspector is installed in '/home/kali/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The script androguard is installed in '/home/kali/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed androguard-4.1.2 apkInspector-1.3.2 banal-1.0.6 dataset-1.6.2 frida-16.6.6 loguru-0.7.3 mtf8-1.0.6

```

Step 2: analyze



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SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
root@kali: /home/kali
File Actions Edit View Help
(kali@kali)~[*]
$ sudo su
[sudo] password for kali:
(root@kali)~[/home/kali]
(root@kali)~[/home/kali]
# androguard analyze /home/kali/Downloads/dr-driving-mod-1.72-drdrivngapk.com.apk
Please be patient, this might take a while.
Found the provided file is of type 'APK'
[INFO ] androguard.apk: Starting analysis on AndroidManifest.xml
[INFO ] androguard.apk: APK file was successfully validated!
[WARNING] androguard.core.api_specific_resources: Requested API level 34 is larger than maximum we have, returning API level 28 instead.
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 07s
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 01s
[INFO ] androguard.analysis: End of creating cross references (XREF) run time: 0min 09s
Added file to session: SHA256::22bfed7fbb20b7d96a5c674147feae66a316585d636ab3e4c73fc63c485bb1ed
Loaded APK file ...
>>> a
<androguard.core.bytecodes.apk.APK object at 0x7f5282150090>
>>> d
[<androguard.core.bytecodes.dvm.DalvikVMFormat object at 0x7f528219c790>, <androguard.core.bytecodes.dvm.DalvikVMFormat object at 0x7f5254a2de50>]
>>> dx
<analysis.Analysis VMs: 2, Classes: 17042, Methods: 86253, Strings: 21794>
Androguard version 3.4.0a1 started
In [1]:
```

```
In [1]: a
Out[1]: <androguard.core.bytecodes.apk.APK at 0x7f5282150090>

In [2]: d
Out[2]:
[<androguard.core.bytecodes.dvm.DalvikVMFormat at 0x7f528219c790>,
 <androguard.core.bytecodes.dvm.DalvikVMFormat at 0x7f5254a2de50>]

In [3]: dx
Out[3]: <analysis.Analysis VMs: 2, Classes: 17042, Methods: 86253, Strings: 21794>

In [4]: a.get_permissions()
Out[4]:
['com.google.android.providers.gsf.permission.READ_GSERVICES',
 'android.permission.ACCESS_NETWORK_STATE',
 'com.android.vending.BILLING',
 'android.permission.ACCESS_AD_SERVICES_ATTRIBUTION',
 'android.permission.ACCESS_AD_SERVICES_TOPICS',
 'android.permission.FOREGROUND_SERVICE',
 'com.ansangha.drdriving.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION',
 'android.permission.INTERNET',
 'android.permission.ACCESS_AD_SERVICES_AD_ID',
 'android.permission.WAKE_LOCK',
 'com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE',
 'com.google.android.gms.permission.AD_ID']

In [5]: a.get_activities()
Out[5]:
['com.ansangha.drdriving.DrDrivingActivity',
 'com.google.android.gms.ads.AdActivity',
 'com.android.billingclient.api.ProxyBillingActivity',
 'com.android.billingclient.api.ProxyBillingActivityV2',
 'com.google.android.gms.auth.api.signin.internal.SignInHubActivity',
 'com.google.android.gms.games.internal.v2.resolution.GamesResolutionActivity',
 'com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity',
 'com.google.android.gms.common.api.GoogleApiActivity',
 'com.google.android.gms.ads.OutOfContextTestingActivity',
 'com.google.android.gms.ads.NotificationHandlerActivity']

In [6]: a.get_package()
Out[6]: 'com.ansangha.drdriving'

In [7]: a.get_app_name()
Out[7]: 'Dr. Driving'

In [8]: a.get_app_icon()
Out[8]: 'res/drawable-xxxhdpi/ic_launcher.png'

In [9]: a.get_androidversion_code()
Out[9]: '72'
```

Step 3: Decompile



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Name: Gargi Sukhatankar

SAP ID: 60019220125

Roll No: B027

Branch: CSE(ICB)

```
root@kali: ~/home/kali
$ androguard decompile -o decompiled_code /home/kali/Downloads/dr-driving-mod-1.72-drdrivingapk.com.apk
[INFO ] androguard.apk: Starting analysis on AndroidManifest.xml
[INFO ] androguard.apk: APK file was successfully validated!
[WARNING] androguard.core.api_specific_resources: Requested API level 34 is larger than maximum we have, returning API level 28 instead.
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 07s
[INFO ] androguard.analysis: Adding DEX file version 35
[INFO ] androguard.analysis: Reading bytecode took : 0min 02s
[INFO ] androguard.analysis: End of creating cross references (XREF) run time: 0min 10s
Dump information /home/kali/Downloads/dr-driving-mod-1.72-drdrivingapk.com.apk in decompiled_code
Create directory decompiled_code
Decompilation ... End
Dump La/a; <init> (V ... source codes ... bytecodes ...
Dump La/a; a (Ljava/lang/StackTraceElement; ... bytecodes ...
Dump La/b; <clinit> (V ... source codes ... bytecodes ...
Dump La/b; b (Ljava/lang/Throwable; Ljava/lang/String;)Ljava/lang/StackTraceElement; ... bytecodes ...
Dump La0/a$a; <init> (La0/a; Lz/e;)V ... source codes ... bytecodes ...
Dump La0/a$a; newCursor (Landroid/database/sqlite/SQLiteDatabase; Landroid/database/sqlite/SQLiteCursorDriver; Ljava/lang/String; Landroid/database/sqlite/SQLiteQuery;)Landroid/database/Cursor; ... bytecodes ...
Dump La0/a$a; <init> (La0/a; Lz/e;)V ... source codes ... bytecodes ...
Dump Lz/b; AB (Lz/e;)Landroid/database/Cursor; ... source codes ... bytecodes ...
Dump Lz/b; F (Ljava/lang/String;)Lz/f; ... bytecodes ...
Dump Lz/b; N (Lz/e; Landroid/os/CancellationSignal;)Landroid/database/Cursor; ... bytecodes ...
Dump Lz/b; Y (Ljava/lang/String; Ljava/lang/Object;)V ... bytecodes ...
Dump Lz/b; a (V ... bytecodes ...
Dump Lz/b; d (V ... bytecodes ...
Dump Lz/b; e0 (Ljava/lang/String;)Landroid/database/Cursor; ... bytecodes ...
Dump Lz/b; getPath ()Ljava/lang/String; ... bytecodes ...
Dump Lz/b; h (V ... bytecodes ...
Dump Lz/b; isOpen ()Z ... bytecodes ...
Dump Lz/b; s0 ()Z ... bytecodes ...
Dump Lz/b; t ()Ljava/util/List; ... bytecodes ...
Dump Lz/b; w (Ljava/lang/String;)V ... bytecodes ...
Dump La0/a; <clinit> (V ... source codes ... bytecodes ...
Dump La0/a; <init> (Landroid/database/sqlite/SQLiteDatabase;)V ... bytecodes ...
```



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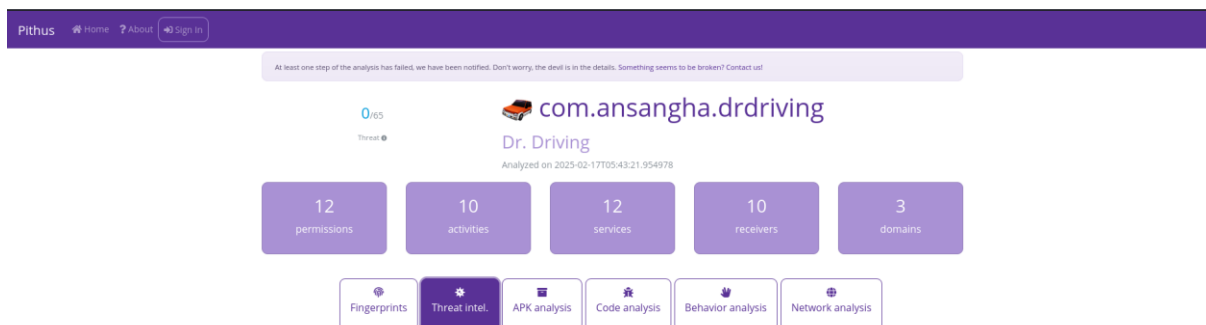
EXPERIMENT 3

Aim: Android Malware analysis using Pithus.

Theory:

Pithus is an open-source, web-based Android malware analysis platform that provides automatic static analysis of APK files to detect potential threats, vulnerabilities, and malicious behavior. It scans APKs for dangerous permissions, suspicious API calls, obfuscation techniques, and embedded payloads, helping security researchers and penetration testers identify malware efficiently. Unlike traditional reverse engineering tools, Pithus requires no installation, making it a fast and accessible solution for Android security analysis. Users can simply upload an APK on [Pithus.org](https://pithus.org) to generate a detailed report, making it a valuable tool for malware detection and forensic investigations.

Screenshots:

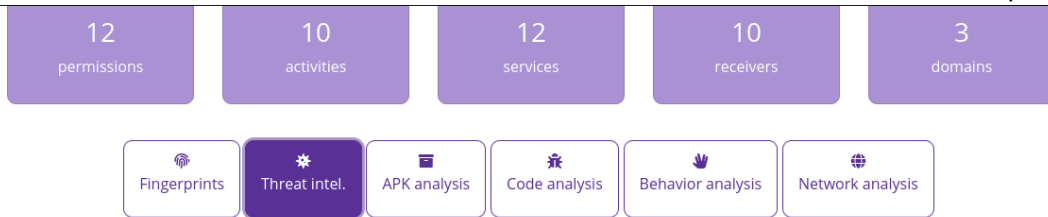


Threats:



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Sample timeline

Certificate valid not before	Jan. 24, 2017, 1:11 p.m.
Oldest file found in APK	Sept. 2, 2024, 3:52 p.m.
Latest file found in APK	Sept. 2, 2024, 3:57 p.m.
First submission on VT	Sept. 2, 2024, 4:19 p.m.
Last submission on VT	Feb. 17, 2025, 12:55 a.m.
Upload on Pithus	Feb. 17, 2025, 5:43 a.m.
Certificate valid not after	June 10, 2044, 1:11 p.m.

VirusTotal

Samples are analysed on upload and are not analysed afterwards again. Make sure to review the status of the analysis on VirusTotal to be sure to have the most up-to-date conclusion.

Score

APK Details:

APK details

Information computed with AndroGuard and Pithus.

Package	com.ansangha.drdriving	
App name	Dr. Driving	
Version name	1.72	
Version code	72	
SDK	21 - 34	
UAID	dfd3e8838174d7c3dce81a65af46d9c3d7d1e65d	
Signature	Signature V1 Signature V2 Signature V3	
Frosting	Not frosted Blocks found within V2 signature: 0x7109871a: Unknown 0xf05368c0: Unknown 0x42726577: Verity padding	

Certificate details

Information computed with AndroGuard.

MD5	3ada5333c68fba2dc7816ead644a47	
SHA1	fd7a95648ff46fb1acab854acfe1bf97ecf92346	



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Certificate details

Information computed with [AndroGuard](#).

MD5	3ada5333c68fba2dcdc7816ead644a47	
SHA1	fd7a95648ff46fb1acab854acfe1bf97ecf92346	
SHA256	1cbd2d0548bb11026d9ab4dbad4a2c9405ce796235d4003bcf0a0aed8bfe0c98	
Issuer	Email Address: a1@tr4oo3r34r34sdfd.com, Common Name: A1, Organizational Unit: A1, Organization: A1, Locality: Lazyland, State/Province: Lazyland, Country: RU	
Not before	2017-01-24T13:11:10+00:00	
Not after	2044-06-10T13:11:10+00:00	

File Analysis

Information computed with [MobSF](#).

Findings	Files
Certificate/Key files hardcoded inside the app.	META-INF/services/io.grpc.ManagedChannelProvider

Manifest analysis

Information computed with [MobSF](#).

Medium	Application Data can be Backed up [android:allowBackup=true] This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
High	Service (com.google.android.gms.auth.api.signin.RevocationBoundService) is Protected by a permission, but the protection level of the permission should be checked. Permission: com.google.android.gms.auth.api.signin.permission.REVOCATION_NOTIFICATION [android:exported=true] A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
High	Activity (com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity) Is not Protected. [android:exported=true] An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
High	Service (androidx.work.impl.background.systemjob.SystemJobService) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.BIND_JOB_SERVICE [android:exported=true] A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
High	Broadcast Receiver (androidx.work.impl.diagnostics.DiagnosticsReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.DUMP [android:exported=true] A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.



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Main Activity

Information computed with AndroGuard.

com.ansangha.drdriving.DrDrivingActivity



Activities

Information computed with AndroGuard.

com.ansangha.drdriving.DrDrivingActivity
com.google.android.gms.ads.AdActivity
com.android.billingclient.api.ProxyBillingActivity
com.android.billingclient.api.ProxyBillingActivityV2
com.google.android.gms.auth.api.signin.internal.SignInHubActivity
com.google.android.gms.games.internal.v2.resolution.GamesResolutionActivity
com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity
com.google.android.gms.common.api.GoogleApiActivity
com.google.android.gms.ads.OutOfContextTestingActivity
com.google.android.gms.ads.NotificationHandlerActivity



Receivers

Information computed with AndroGuard.

com.google.android.datatransport.runtime.scheduling.jobscheduling.AlarmManagerSchedulerBroadcastReceiver
com.google.android.gms.measurement.AppMeasurementReceiver
androidx.work.impl.utils.ForceStopRunnable\$BroadcastReceiver
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryChargingProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$StorageNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$NetworkStateProxy
androidx.work.impl.background.systemalarm.RescheduleReceiver



Receivers

Information computed with AndroGuard.

com.google.android.datatransport.runtime.scheduling.jobscheduling.AlarmManagerSchedulerBroadcastReceiver
com.google.android.gms.measurement.AppMeasurementReceiver
androidx.work.impl.utils.ForceStopRunnable\$BroadcastReceiver
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryChargingProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$StorageNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$NetworkStateProxy
androidx.work.impl.background.systemalarm.RescheduleReceiver
androidx.work.impl.background.systemalarm.ConstraintProxyUpdateReceiver
androidx.work.impl.diagnostics.DiagnosticsReceiver

Services

Information computed with AndroGuard.

com.google.android.gms.auth.api.signin.RevocationBoundService
com.google.firebase.components.ComponentDiscoveryService
com.google.firebase.sessions.SessionLifecycleService
com.google.android.datatransport.runtime.backends.TransportBackendDiscovery
com.google.android.datatransport.runtime.scheduling.jobscheduling.JobInfoSchedulerService
com.google.android.gms.measurement.AppMeasurementService
com.google.android.gms.measurement.AppMeasurementJobService
com.google.android.gms.ads.AdService
androidx.work.impl.background.systemalarm.SystemAlarmService
androidx.work.impl.background.systemjob.SystemJobService
androidx.work.impl.foreground.SystemForegroundService
androidx.room.MultiInstanceInvalidationService



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Code Analysis:

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NIAP analysis

Information computed with MobSF.

FCS_RBG_EXT.1.1	The application invoke platform-provided DRBG functionality for its cryptographic operations. Random Bit Generation Services
FCS_STO_EXT.1.1	The application does not store any credentials to non-volatile memory. Storage of Credentials
FCS_CKM_EXT.1.1	The application implement asymmetric key generation. Cryptographic Key Generation Services
FDP_DEC_EXT.1.1	The application has access to [network connectivity]. Access to Platform Resources
FDP_DEC_EXT.1.2	The application has access to no sensitive information repositories. Access to Platform Resources
FDP_NET_EXT.1.1	The application has user/application initiated network communications. Network Communications
FDP_DAR_EXT.1.1	The application implement functionality to encrypt sensitive data in non-volatile memory. Encryption Of Sensitive Application Data
FMT_MEC_EXT.1.1	The application invoke the mechanisms recommended by the platform vendor for storing and setting configuration options. Supported Configuration Mechanism
FTP_DIT_EXT.1.1	The application does encrypt some transmitted data with HTTPS/TLS/SSH between itself and another trusted IT product. Protection of Data in Transit
FCS_RBG_EXT.2.1 FCS_RBG_EXT.2.2	The application perform all deterministic random bit generation (DRBG) services in accordance with NIST Special Publication 800-90A using Hash_DRBG. The deterministic RBG is seeded by an entropy source that accumulates entropy from a platform-based DRBG and a software-based noise source, with a minimum of 256 bits of entropy at least equal to the greatest security strength (according to NIST SP 800-57) of the keys and hashes that it will generate. Random Bit Generation from Application
FCS_COP.1.1(1)	The application perform encryption/decryption in accordance with a specified cryptographic algorithm AES-CBC (as defined in NIST SP 800-38A) mode or AES-GCM (as defined in NIST SP 800-38D) and cryptographic key sizes 256-bit/128-bit.

Code analysis

Information computed with MobSF.

Medium CVSS:7.5 	🔗 The App uses an Insecure Random Number Generator. MASVS: MSTG-CRYPTO-6 CWE-330 Use of Insufficiently Random Values M5: Insufficient Cryptography
Medium CVSS:7.4 	🔗 Files may contain hardcoded sensitive information like usernames, passwords, keys etc. MASVS: MSTG-STORAGE-14 CWE-312 Cleartext Storage of Sensitive Information M9: Reverse Engineering
Low CVSS:7.5 	🔗 The App logs information. Sensitive information should never be logged. MASVS: MSTG-STORAGE-3 CWE-532 Insertion of Sensitive Information into Log File
Medium CVSS:5.9 	🔗 App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database. CWE-89 Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') M7: Client Code Quality
Medium CVSS:5.9 	🔗 SHA-1 is a weak hash known to have hash collisions. MASVS: MSTG-CRYPTO-4 CWE-327 Use of a Broken or Risky Cryptographic Algorithm M5: Insufficient Cryptography
High CVSS:5.5 	🔗 App can read/write to External Storage. Any App can read data written to External Storage. MASVS: MSTG-STORAGE-2 CWE-276 Incorrect Default Permissions M2: Insecure Data Storage

Behaviour Analysis:



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Permissions analysis

Information computed with MobSF.

Low	android.permission.INTERNET	full Internet access Allows an application to create network sockets.	
Low	android.permission.ACCESS_NETWORK_STATE	view network status Allows an application to view the status of all networks.	
Low	android.permission.WAKE_LOCK	prevent phone from sleeping Allows an application to prevent the phone from going to sleep.	
Low	android.permission.FOREGROUND_SERVICE	Allows a regular application to use Service.startForeground.	
	com.android.vending.BILLING	Unknown permission Unknown permission from android reference	
	com.google.android.providers.gsf.permission.READ_GSERVICES	Unknown permission Unknown permission from android reference	
	com.google.android.gms.permission.AD_ID	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_ATTRIBUTION	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_AD_ID	Unknown permission Unknown permission from android reference	
	com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_TOPICS	Unknown permission Unknown permission from android reference	
	com.ansangha.drdriving.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	Unknown permission Unknown permission from android reference	



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Tracking analysis

Information computed with Exodus-core.

Google AdMob	https://reports.exodus-privacy.eu.org/fr/trackers/312
Google CrashLytics	https://reports.exodus-privacy.eu.org/fr/trackers/27
Google Firebase Analytics	https://reports.exodus-privacy.eu.org/fr/trackers/49

Threat analysis

Information computed with Quark-Engine.

Confidence: 100%	Check if the network is connected
Confidence: 100%	Load external class
Confidence: 100%	Query the current data network type
Confidence: 100%	Implicit intent(view a web page, make a phone call, etc.)
Confidence: 100%	Find a method from given class name, usually for reflection
Confidence: 100%	Check the active network type
Confidence: 100%	Read data and put it into a buffer stream
Confidence: 100%	Put the compressed bitmap data into JSON object
Confidence: 100%	Method reflection
Confidence:	Connect to a URL and read data from it



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Behavior analysis

Information computed with MobSF.

Base64 decode
Base64 encode
Certificate handling
Get system service
Http connection
Https connection
Inter process communication
Java reflection
Local file i/o operations
Message digest
Starting activity
Starting service
Tcp socket

Network Analysis:

Domains analysis

Information computed with MobSF.

US	crucial-bonsai-821.firebaseio.com	🔍 📄 G	34.120.206.254	📄 G
US	firebase.google.com	🔍 📄 G	142.250.185.206	📄 G
US	github.com	🔍 📄 G	140.82.121.3	📄 G

URL analysis

Information computed with MobSF.

https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/crashlytics/get-started?platform=android#add-plugin Defined in x1/q.java	📄
https://crucial-bonsai-821.firebaseio.com Defined in Android String Resource	📄



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Conclusion:

Thus we successfully scanned Dr. Driving game APK for malware using Pithus on web browser in kali. It provided us with a detailed report consisting of fingerprints, threats, domains, urls, analysis, code analysis and many more things.



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EXPERIMENT 4

Aim: To analyze original and mod APK of using Quark.

Theory:

What is Quark

Quark is a static analysis tool designed to detect malicious activities and vulnerabilities in Android applications by scanning APK files. It operates using a rule-based detection system that identifies suspicious patterns in an app's code without executing it. Unlike dynamic analysis tools, Quark does not require a runtime environment, making it a fast and lightweight option for security researchers and penetration testers. It specializes in uncovering privacy leaks, backdoors, spyware, and privilege escalation risks, making it a valuable asset in Android malware analysis and security auditing.

APK Analysis using Quark

APK analysis using Quark involves scanning an application's code for predefined malicious signatures and behavior patterns. The tool inspects API calls, permissions, and embedded scripts to flag potential security threats. The output provides detailed insights into detected risks, such as unauthorized data collection, suspicious network activity, and privilege abuse. Since it focuses on static code analysis, Quark is best used in combination with other tools like MobSF for dynamic testing and JADX for decompilation to gain a more complete security assessment of an Android application.

Output:

- 1) Quark Tool



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```
(root@kali)-[/home/kali/Downloads]
# quark --help

An Obfuscation-Neglect Android Malware Scoring System

Usage: quark [OPTIONS]

Quark is an Obfuscation-Neglect Android Malware Scoring System

Options:
  -s, --summary TEXT          Show summary report. Optionally specify the
                              name of a rule/label
  -d, --detail TEXT           Show detail report. Optionally specify the
                              name of a rule/label
  -o, --output FILE           Output report in JSON
  -w, --webreport FILE        Generate web report
  -a, --apk FILE              APK file [required]
  -r, --rule PATH             Rules directory [default: /root/.quark-
                              engine/quark-rules/rules]
  -g, --graph [png|json]     Create call graph to call_graph_image
                              directory
  -c, --classification        Show rules classification
  -t, --threshold [100|80|60|40|20]
                              Set the lower limit of the confidence
                              threshold
  -i, --list [all|native|custom]
                              List classes, methods and descriptors
  -p, --permission            List Android permissions
  -l, --label [max|detailed]  Show report based on label of rules
  -C, --comparison            Behaviors comparison based on max confidence
                              of rule labels
  --generate-rule DIRECTORY  Generate rules and output to given directory
  --core-library [androguard|rizin]
                              Specify the core library used to analyze an
                              APK
  --multi-process INTEGER RANGE
                              Allow analyzing APK with N processes, where
                              N doesn't exceeds the number of usable CPUs
                              - 1 to avoid memory exhaustion. [x ≥ 1]
  --version                  Show the version and exit.
  --help                    Show this message and exit.
```

2) APK Analysis

```
(kali@kali)-[/Downloads]
$ quark -s flashlight-1-2-3.apk -s

An Obfuscation-Neglect Android Malware Scoring System

100% [!] WARNING: Moderate Risk [4] Total Score: 211 211/211 [00:55:00:00, 3.80it/s]

File Name | Rule | Confidence | Score | Weight |
|---|---|---|---|---|
| 00064.json | Monitor incoming call status | 40% | 1 | 0.125 |
| 00211.json | Open an SMS in Webview | 20% | 1 | 0.0025 |
| 00184.json | Set camera preview texture | 40% | 1 | 0.125 |
| 00180.json | Load native libraries(.so) via System.loadLibrary (60% means caught) | 20% | 1 | 0.0025 |
| 00079.json | Hide the current app's icon | 40% | 1 | 0.125 |
| 00136.json | Stop recording | 40% | 1 | 0.125 |
| 00054.json | Install other APKs from file | 40% | 1 | 0.125 |
| 00045.json | Query the name of currently running application | 20% | 1 | 0.0025 |
| 00088.json | Create a secure socket connection to the given host address | 60% | 1 | 0.25 |
| 00123.json | Save the response to JSON after connecting to the remote server | 40% | 1 | 0.125 |
| 00067.json | Check the current network type | 40% | 1 | 0.125 |
| 00097.json | Get the sender address of the SMS and put it into JSON | 40% | 1 | 0.125 |
| 00016.json | Get JSON object prepared and fill in location info | 40% | 1 | 0.125 |
| 00202.json | Make a phone call | 100% | 1 | 1 |
| 00157.json | Instantiate new object using reflection, possibly used for dexClassLoader | 100% | 1 | 1 |
| 00034.json | Query the current data network type | 100% | 1 | 1 |
| 00053.json | Implicit intent(view a web page, make a phone call, etc.) via setData | 100% | 1 | 1 |
| 00199.json | Stop recording and release recording resources | 20% | 1 | 0.0025 |
| 00089.json | Connect to a URL and receive input stream from the server | 100% | 1 | 1 |
| 00163.json | Perform accessibility service action on accessibility node info | 20% | 1 | 0.0025 |
| 00185.json | Start capturing camera preview frames to the screen | 60% | 1 | 0.25 |
| 00173.json | Get bounds in screen of an AccessibilityNodeInfo and perform action | 40% | 1 | 0.125 |
| 00124.json | Check the current active network type | 40% | 1 | 0.125 |
| 00197.json | Set the audio encoder and initialize the recorder | 20% | 1 | 0.0025 |
| 00062.json | Get the current WiFi MAC address | 40% | 1 | 0.125 |
| 00066.json | Query the ICCID number | 40% | 1 | 0.125 |
| 00110.json | Query the ICCID number | 20% | 1 | 0.0025 |
| 00076.json | Get the network operator name | 100% | 1 | 1 |
| 00200.json | Capture the contents of the device screen | 20% | 1 | 0.0025 |
| 00077.json | Read sensitive data(SMS, CALLLOG, etc) | 100% | 1 | 1 |
| 00016.json | Get location info of the device and put it to JSON object | 100% | 1 | 1 |
```

3) Permissions



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Branch: CSE(ICB)

```
(kali@kali)-[~/Downloads]
$ quark -a flashlight-1-2-3.apk -p

An Obfuscation-Neglect Android Malware Scoring System

com.google.android.c2dm.permission.RECEIVE
android.permission.WAKE_LOCK
com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE
com.surpax.ledflashlight.panel.permission.C2D_MESSAGE
android.permission.ACCESS_NETWORK_STATE
android.permission.FLASHLIGHT
com.surpax.ledflashlight.panel.permission.FRAMEWORK_SECURITY
android.permission.INTERNET
```

4) MOD APK Analysis

```
(root@kali)-[/home/kali/Downloads]
$ quark -a Flashlight\ Plus\ v2.8.3\ \Plus\).apk -s

An Obfuscation-Neglect Android Malware Scoring System

Requested API level 33 is larger than maximum we have, returning API level 28 instead.
100%
[!] WARNING: Moderate Risk
[*] Total Score: 211

+-----+-----+-----+-----+-----+
| Filename | Rule | Confidence | Score | Weight |
+-----+-----+-----+-----+-----+
| 00064.json | Monitor incoming call status | 40% | 1 | 0.125 |
| 00211.json | Open an URL in Webview | 20% | 1 | 0.0625 |
| 00184.json | Set camera preview texture | 60% | 1 | 0.25 |
| 00180.json | Load native libraries(.so) via System.loadLibrary (60% means caught) | 60% | 1 | 0.25 |
| 00079.json | Hide the current app's icon | 100% | 1 | 1.0 |
| 00136.json | Stop recording | 20% | 1 | 0.0625 |
| 00054.json | Install other APKs from file | 40% | 1 | 0.125 |
| 00045.json | Query the name of currently running application | 20% | 1 | 0.0625 |
| 00088.json | Create a secure socket connection to the given host address | 20% | 1 | 0.0625 |
| 00123.json | Save the response to JSON after connecting to the remote server | 40% | 1 | 0.125 |
| 00097.json | Check the current network type | 100% | 1 | 1.0 |
| 00097.json | Get the sender address of the SMS and put it into JSON | 40% | 1 | 0.125 |
| 00018.json | Get JSON object prepared and fill in location info | 40% | 1 | 0.125 |
| 00202.json | Make a phone call | 100% | 1 | 1.0 |
| 00157.json | Instantiate new object using reflection, possibly used for dexClassLoader | 100% | 1 | 1.0 |
```

Conclusion:

Thus we successfully analyzed original and mod APK of using Quark. Quark effectively detects static threats, combining it with dynamic analysis and reverse engineering tools can provide a more comprehensive security assessment, making it a crucial step in evaluating APK authenticity and safety.



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EXPERIMENT NO. 5

Aim: To implement SQL lite browser to view data on android.

SQLite Browser: SQLite Browser (also known as DB Browser for SQLite) is a free, open-source, visual tool designed to simplify working with SQLite database files. It provides an intuitive graphical interface that allows users to create, browse, edit, and run SQL queries on SQLite databases without writing code. Ideal for developers and testers, it makes inspecting and managing local app data straightforward, especially during mobile app development or debugging.

Analysis:

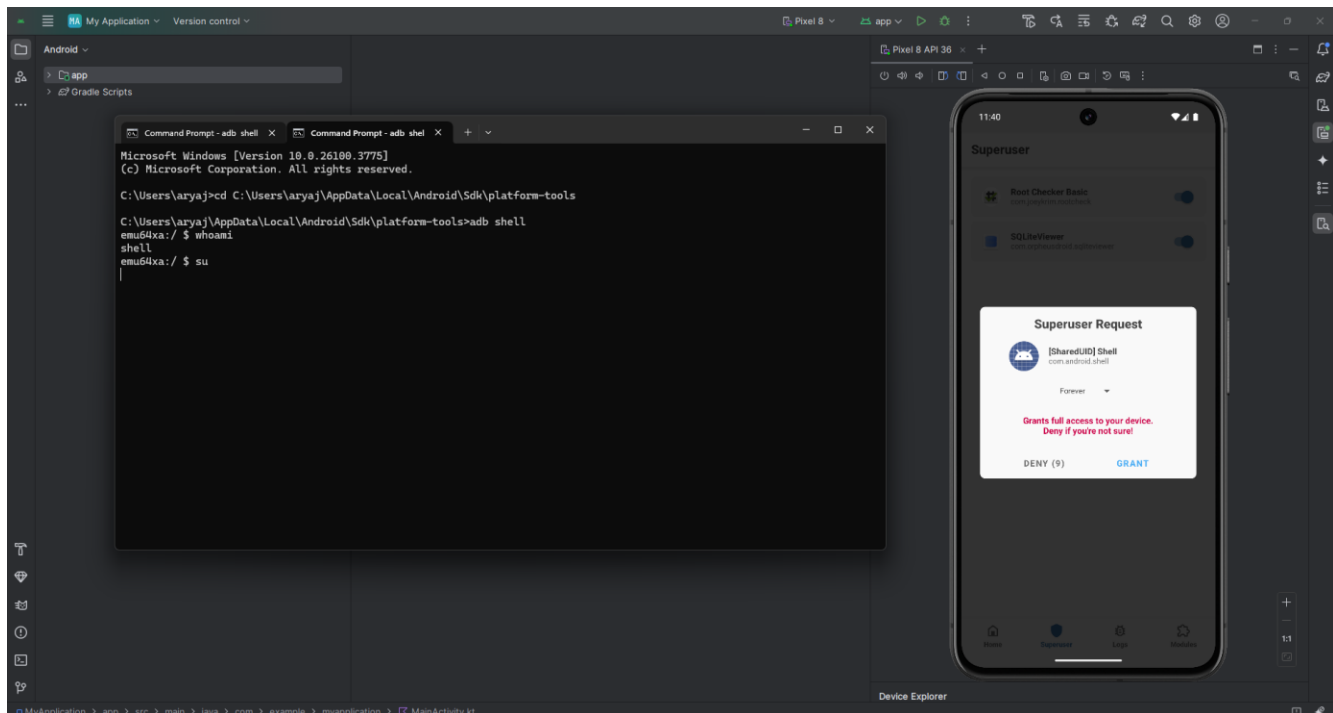


Figure 1: Granting Root Permissions



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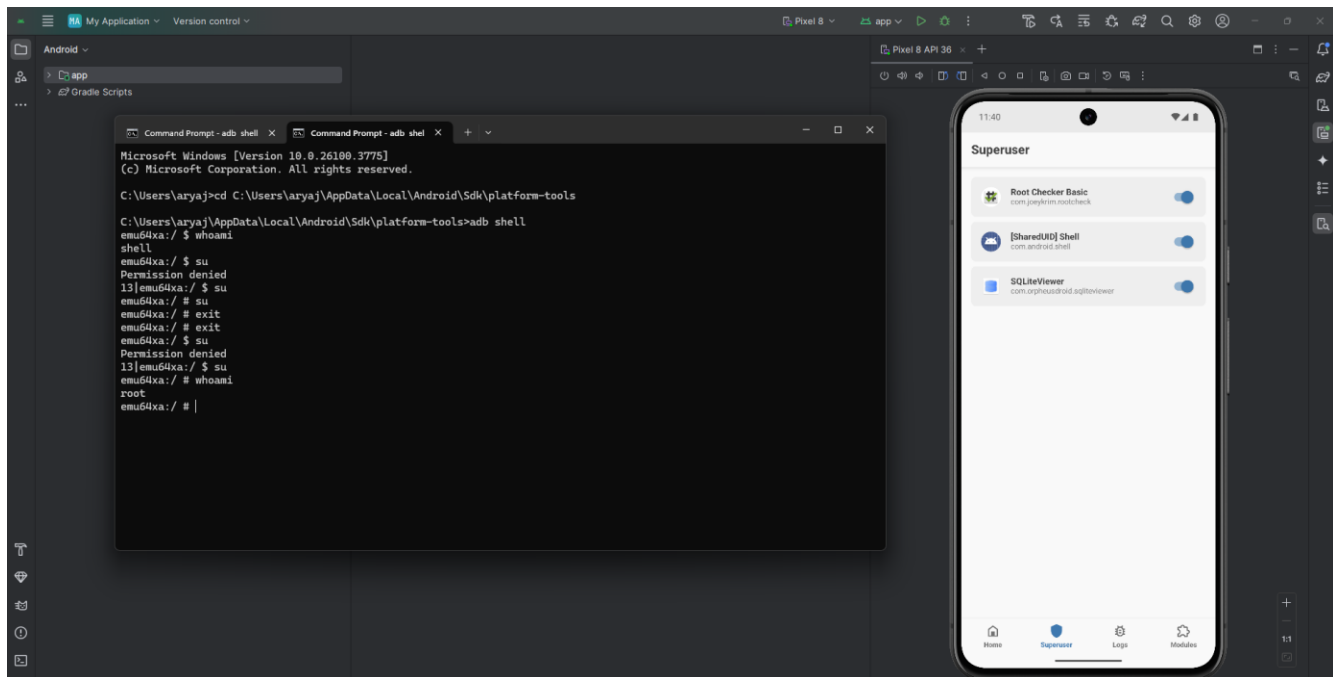


Figure 2: Successfully Granted Root Permissions



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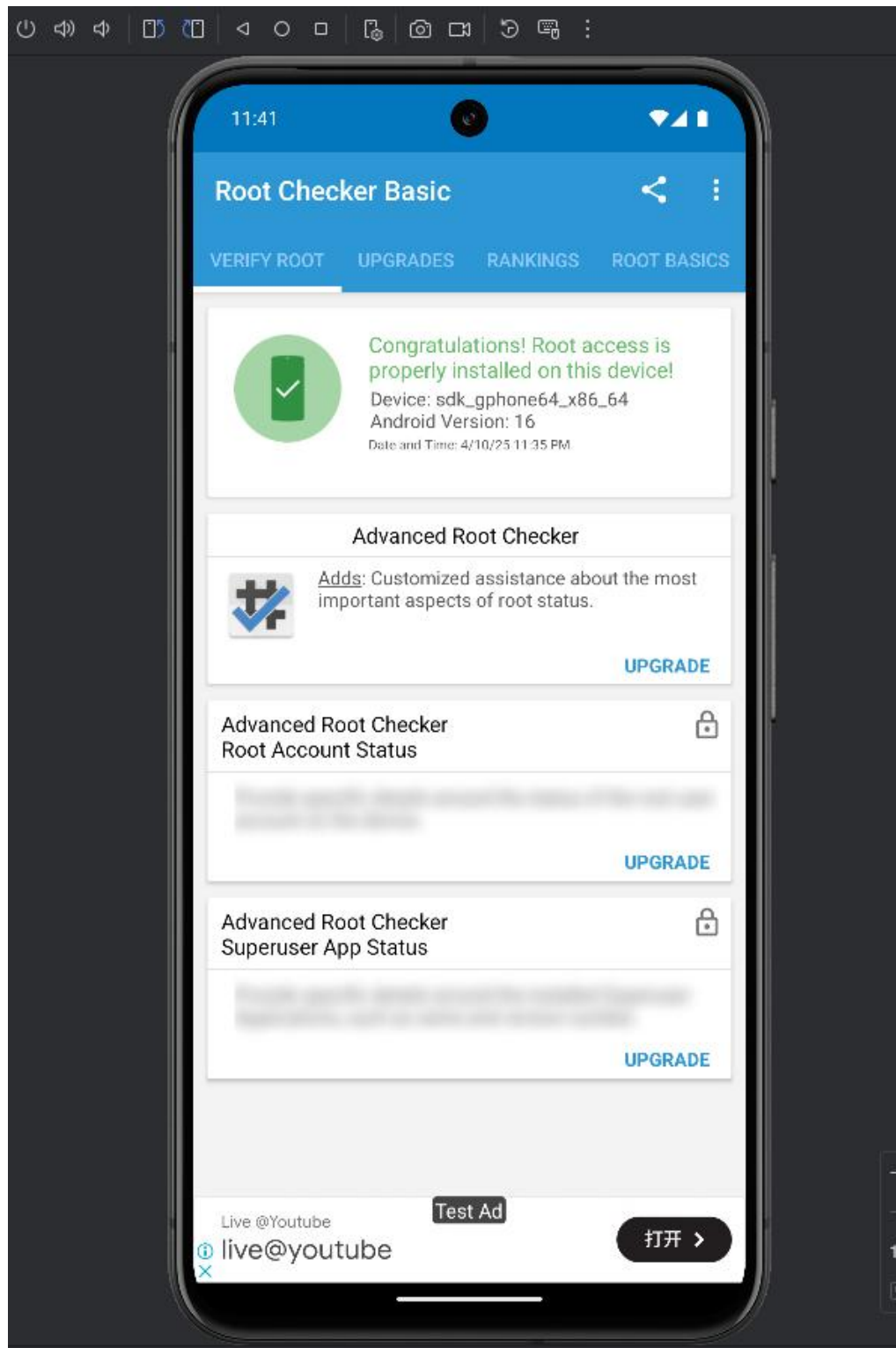


Figure 3: Root Checker



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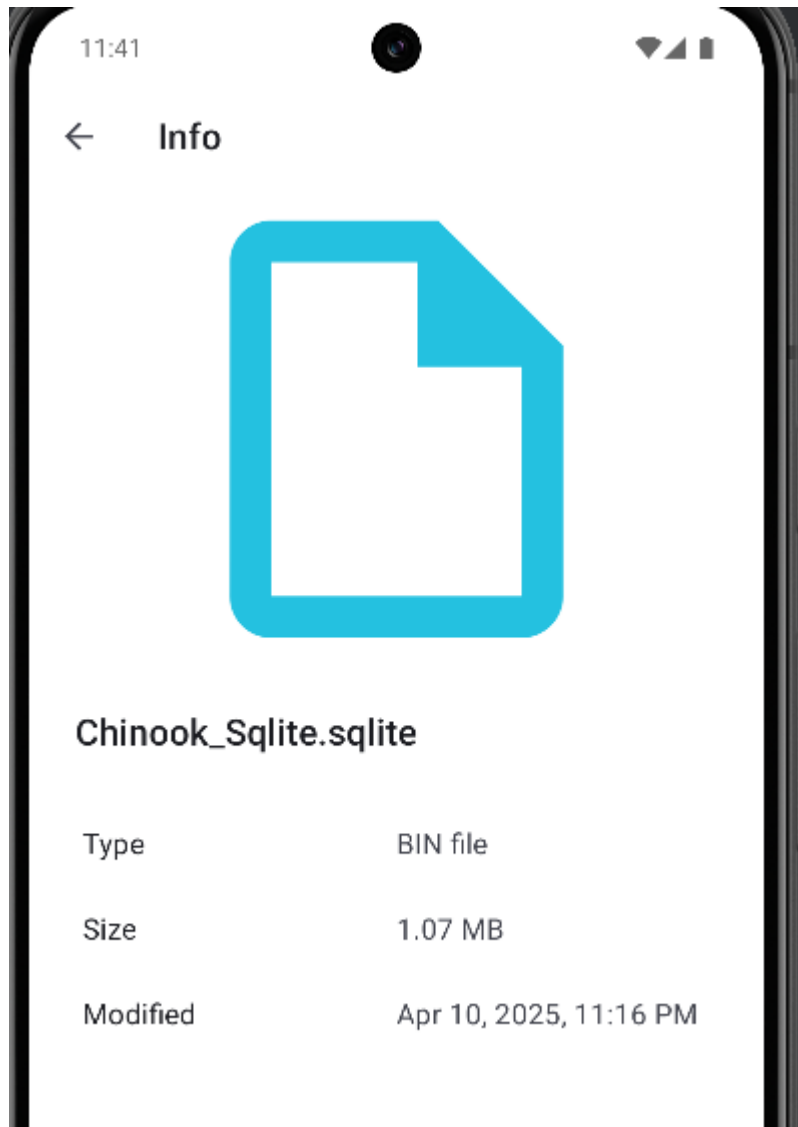


Figure 3: SQLite File created



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Name: Gargi Sukhatankar
Roll No: B027

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Branch: CSE(ICB)

D8 Browser for SQLite - C:\Users\arya\Downloads\Chinook_Sqlite.sqlite

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pgmas Execute SQL

Create Table Create Index Modify Table Delete Table Print Refresh

Name	Type	Schema
Tables (11)		
Album		CREATE TABLE [Album] ([AlbumId] INTEGER NOT NULL, [Title] NVARCHAR(160) NOT NULL, [ArtistId] INTEGER NOT NULL, CONSTRAINT [PK_Album] PRIMARY KEY ([AlbumId]), FOREIGN KEY ([ArtistId]) REFER
Artist		CREATE TABLE [Artist] ([ArtistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Artist] PRIMARY KEY ([ArtistId]))
Customer		CREATE TABLE [Customer] ([CustomerId] INTEGER NOT NULL, [FirstName] NVARCHAR(40) NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [Company] NVARCHAR(80), [Address] NVARCHAR(70), [City] NVA
Employee		CREATE TABLE [Employee] ([EmployeeId] INTEGER NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [FirstName] NVARCHAR(20) NOT NULL, [Title] NVARCHAR(30), [ReportsTo] INTEGER, [BirthDate] DATETI
Genre		CREATE TABLE [Genre] ([GenreId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Genre] PRIMARY KEY ([GenreId]))
Invoice		CREATE TABLE [Invoice] ([InvoiceId] INTEGER NOT NULL, [CustomerId] INTEGER NOT NULL, [InvoiceDate] DATETIME NOT NULL, [BillingAddress] NVARCHAR(70), [BillingCity] NVARCHAR(40), [BillingState] NVA
InvoiceLine		CREATE TABLE [InvoiceLine] ([InvoiceLineId] INTEGER NOT NULL, [InvoiceId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, [UnitPrice] NUMERIC(10,2) NOT NULL, [Quantity] INTEGER NOT NULL, CONSTRA
MediaType		CREATE TABLE [MediaType] ([MediaTypeId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_MediaType] PRIMARY KEY ([MediaTypeId]))
Playlist		CREATE TABLE [Playlist] ([PlaylistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Playlist] PRIMARY KEY ([PlaylistId]))
PlaylistTrack		CREATE TABLE [PlaylistTrack] ([PlaylistId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, CONSTRAINT [PK_PlaylistTrack] PRIMARY KEY ([PlaylistId], [TrackId]), FOREIGN KEY ([PlaylistId]) REFERENCES [Pl
Track		CREATE TABLE [Track] ([TrackId] INTEGER NOT NULL, [Name] NVARCHAR(200) NOT NULL, [AlbumId] INTEGER, [MediaTypeId] INTEGER NOT NULL, [GenreId] INTEGER, [Composer] NVARCHAR(220), [Mills
Indices (21)		
IFK_AlbumArtistId		CREATE INDEX [IFK_AlbumArtistId] ON [Album] ([ArtistId])
IFK_CustomerSupportRepId		CREATE INDEX [IFK_CustomerSupportRepId] ON [Customer] ([SupportRepId])
IFK_EmployeeReportsTo		CREATE INDEX [IFK_EmployeeReportsTo] ON [Employee] ([ReportsTo])
IFK_InvoiceCustomerId		CREATE INDEX [IFK_InvoiceCustomerId] ON [Invoice] ([CustomerId])
IFK_InvoiceLineInvoiceId		CREATE INDEX [IFK_InvoiceLineInvoiceId] ON [InvoiceLine] ([InvoiceId])
IFK_InvoiceLineTrackId		CREATE INDEX [IFK_InvoiceLineTrackId] ON [InvoiceLine] ([TrackId])
IFK_PlaylistTrackTrackId		CREATE INDEX [IFK_PlaylistTrackTrackId] ON [PlaylistTrack] ([TrackId])
IFK_TrackAlbumId		CREATE INDEX [IFK_TrackAlbumId] ON [Track] ([AlbumId])
IFK_TrackGenreId		CREATE INDEX [IFK_TrackGenreId] ON [Track] ([GenreId])
IFK_TrackMediaTypeId		CREATE INDEX [IFK_TrackMediaTypeId] ON [Track] ([MediaTypeId])
IFK_Album		CREATE UNIQUE INDEX [IFK_Album] ON [Album]([AlbumId])
IFK_Artist		CREATE UNIQUE INDEX [IFK_Artist] ON [Artist]([ArtistId])
IFK_Customer		CREATE UNIQUE INDEX [IFK_Customer] ON [Customer]([CustomerId])
IFK_Employee		CREATE UNIQUE INDEX [IFK_Employee] ON [Employee]([EmployeeId])
IFK_Genre		CREATE UNIQUE INDEX [IFK_Genre] ON [Genre]([GenreId])
IFK_Invoice		CREATE UNIQUE INDEX [IFK_Invoice] ON [Invoice]([InvoiceId])
IFK_InvoiceLine		CREATE UNIQUE INDEX [IFK_InvoiceLine] ON [InvoiceLine]([InvoiceLineId])
IFK_MediaType		CREATE UNIQUE INDEX [IFK_MediaType] ON [MediaType]([MediaTypeId])
IFK_Playlist		CREATE UNIQUE INDEX [IFK_Playlist] ON [Playlist]([PlaylistId])
IFK_PlaylistTrack		CREATE UNIQUE INDEX [IFK_PlaylistTrack] ON [PlaylistTrack]([PlaylistId], [TrackId])
IFK_Track		CREATE UNIQUE INDEX [IFK_Track] ON [Track]([TrackId])
Views (0)		
Triggers (0)		

Name	Type	Schema
Tables (11)		
Album		CREATE TABLE [Album] ([AlbumId] INTEGER NOT NULL, [Title] NVARCHAR(160) NOT NULL, [ArtistId] INTEGER NOT NULL, CONSTRAINT [PK_Album] PRIMARY KEY ([AlbumId]), FOREIGN KEY ([ArtistId]) REF
AlbumId	INTEGER	"AlbumId" INTEGER NOT NULL
Title	NVARCHAR(1...	"Title" NVARCHAR(160) NOT NULL
ArtistId	INTEGER	"ArtistId" INTEGER NOT NULL
Artist		CREATE TABLE [Artist] ([ArtistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Artist] PRIMARY KEY ([ArtistId]))
ArtistId	INTEGER	"ArtistId" INTEGER NOT NULL
Name	NVARCHAR(1...	"Name" NVARCHAR(120)
Customer		CREATE TABLE [Customer] ([CustomerId] INTEGER NOT NULL, [FirstName] NVARCHAR(40) NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [Company] NVARCHAR(80), [Address] NVARCHAR(70), [City] N
CustomerId	INTEGER	"CustomerId" INTEGER NOT NULL
FirstName	NVARCHAR(40)	"FirstName" NVARCHAR(40) NOT NULL
LastName	NVARCHAR(20)	"LastName" NVARCHAR(20) NOT NULL
Company	NVARCHAR(80)	"Company" NVARCHAR(80)
Address	NVARCHAR(70)	"Address" NVARCHAR(70)
City	NVARCHAR(40)	"City" NVARCHAR(40)
State	NVARCHAR(40)	"State" NVARCHAR(40)
Country	NVARCHAR(40)	"Country" NVARCHAR(40)
PostalCode	NVARCHAR(10)	"PostalCode" NVARCHAR(10)
Phone	NVARCHAR(24)	"Phone" NVARCHAR(24)
Fax	NVARCHAR(24)	"Fax" NVARCHAR(24)
Email	NVARCHAR(60)	"Email" NVARCHAR(60) NOT NULL
SupportRepId	INTEGER	"SupportRepId" INTEGER
Employee		CREATE TABLE [Employee] ([EmployeeId] INTEGER NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [FirstName] NVARCHAR(20) NOT NULL, [Title] NVARCHAR(30), [ReportsTo] INTEGER, [BirthDate] DATE
Genre		CREATE TABLE [Genre] ([GenreId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Genre] PRIMARY KEY ([GenreId]))
Invoice		CREATE TABLE [Invoice] ([InvoiceId] INTEGER NOT NULL, [CustomerId] INTEGER NOT NULL, [InvoiceDate] DATETIME NOT NULL, [BillingAddress] NVARCHAR(70), [BillingCity] NVARCHAR(40), [BillingState] N
InvoiceId	INTEGER	"InvoiceId" INTEGER NOT NULL
CustomerId	INTEGER	"CustomerId" INTEGER NOT NULL
InvoiceDate	DATETIME	"InvoiceDate" DATETIME NOT NULL
BillingAddress	NVARCHAR(70)	"BillingAddress" NVARCHAR(70)
BillingCity	NVARCHAR(40)	"BillingCity" NVARCHAR(40)
BillingState	NVARCHAR(40)	"BillingState" NVARCHAR(40)
BillingCountry	NVARCHAR(40)	"BillingCountry" NVARCHAR(40)
BillingPostalCode	NVARCHAR(10)	"BillingPostalCode" NVARCHAR(10)
Total	NUMERIC(10, 2)	"Total" NUMERIC(10, 2) NOT NULL
InvoiceLine		CREATE TABLE [InvoiceLine] ([InvoiceLineId] INTEGER NOT NULL, [InvoiceId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, [UnitPrice] NUMERIC(10,2) NOT NULL, [Quantity] INTEGER NOT NULL, CONST
MediaType		CREATE TABLE [MediaType] ([MediaTypeId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_MediaType] PRIMARY KEY ([MediaTypeId]))
Playlist		CREATE TABLE [Playlist] ([PlaylistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Playlist] PRIMARY KEY ([PlaylistId]))
PlaylistTrack		CREATE TABLE [PlaylistTrack] ([PlaylistId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, CONSTRAINT [PK_PlaylistTrack] PRIMARY KEY ([PlaylistId], [TrackId]), FOREIGN KEY ([PlaylistId]) REFERENCES
Track		CREATE TABLE [Track] ([TrackId] INTEGER NOT NULL, [Name] NVARCHAR(200) NOT NULL, [AlbumId] INTEGER, [MediaTypeId] INTEGER NOT NULL, [GenreId] INTEGER, [Composer] NVARCHAR(220), [Mills
Indices (21)		
IFK_AlbumArtistId		CREATE INDEX [IFK_AlbumArtistId] ON [Album] ([ArtistId])

Figure 5: Reading SQLite File using SQLite Browser



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Conclusion:

Successfully Rooted Android Emulator and used SQLite browser to view database file for Android Apk.



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EXPERIMENT 6

Aim: To perform analysis of Android apps using android lint.

Theory:

The Android Lint command is a tool provided by the Android SDK that helps developers identify and correct issues in their Android projects. It performs static analysis of the code, looking for potential bugs, performance issues, usability problems, accessibility issues, and more. Lint checks can range from simple syntax errors to more complex issues that might not be immediately obvious, such as deprecated API usage, missing translations, or potential security vulnerabilities. The Android Lint tool is integrated into the Android Studio IDE, but it can also be run from the command line. This makes it accessible to developers who prefer working outside of an IDE or who need to integrate lint checks into continuous integration (CI) pipelines. When you run the Android Lint command, it scans your project's source code and resources, generating a report that lists any issues it finds. The report can be viewed in Android Studio, or it can be exported to various formats, such as HTML or XML, for further analysis or documentation.

Code:

Execute `.\gradlew lint` in the terminal of the Android Studio

```
PS C:\Users\djsce.student\AndroidStudioProjects> .\gradlew lint

Welcome to Gradle 8.2!

Here are the highlights of this release:
- Kotlin DSL: new reference documentation, assignment syntax by default
- Kotlin DSL is now the default with Gradle init
> Task :app:lintReportDebug
Wrote HTML report to file:///C:/Users/djsce.student/AndroidStudioProjects/app/build/reports/lint-results-debug.html

BUILD SUCCESSFUL in 48s
23 actionable tasks: 12 executed, 11 up-to-date
PS C:\Users\djsce.student\AndroidStudioProjects> 
```

Open Lint Report in Browser



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Lint Report: 34 warnings

Check performed at Mon Apr 21 11:16:26 IST 2025 by AGP (8.2.1)

Overview

Correctness
1 ScopedStorage : Affected by scoped storage
1 FragmentTagUsage : Use FragmentContainerView instead of the <fragment> tag
1 ManifestOrder : Incorrect order of elements in manifest
1 RedundantLabel : Redundant label on activity
7 GradleDependency : Obsolete Gradle Dependency

Performance
1 ObsoleteSdkInt : Obsolete SDK_INT Version Check
9 UnusedResources : Unused resources

Usability
4 TextFields : Missing InputType
4 Autofill : Use Autofill

Internationalization
5 HardcodedText : Hardcoded text

Included Additional Checks (36)

Disabled Checks (39)

DISMISS

Lint Report: 34 warnings

Check performed at Mon Apr 21 11:16:26 IST 2025 by AGP (8.2.1)

Affected by scoped storage

[./src/main/AndroidManifest.xml:28](#): READ_EXTERNAL_STORAGE is deprecated (and is not granted) when targeting Android 13+. If you need to query or interact with MediaStore or media files on the shared storage, you should instead use one or more new storage permissions: READ_MEDIA_IMAGES, READ_MEDIA_VIDEO or READ_MEDIA_AUDIO.

```
25 </activity>
26 </application>
27 <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
28 </manifest>
```

ScopedStorage Correctness Warning Priority 8/10

EXPLAIN DISMISS

Use FragmentContainerView instead of the <fragment> tag

[./src/main/res/layout/content_main.xml:8](#): Replace the <fragment> tag with FragmentContainerView

```
5 android:layout_height="match_parent"
6 app:layout_behavior="getString/appbar_scrolling_view_behavior">
7 <fragment
8   android:id="@+id/new_host_fragment_content_main"
9   android:name="androidx.navigation.fragment.NavHostFragment"
10   android:layout_width="wrap">
```

Vendor: Android Open Source Project
Identifier: androidx.fragment
Feedback: <https://issuetracker.google.com/issues/new?component=600964>

FragmentTagUsage Correctness Warning Priority 5/10

EXPLAIN DISMISS

Lint Report: 34 warnings

Check performed at Mon Apr 21 11:16:26 IST 2025 by AGP (8.2.1)

EXPLAIN DISMISS

Incorrect order of elements in manifest

[./src/main/AndroidManifest.xml:28](#): <uses-permission> tag appears after <application> tag

```
25 </activity>
26 </application>
27 <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
28 </manifest>
```

ManifestOrder Correctness Warning Priority 5/10

EXPLAIN DISMISS

Redundant label on activity

[./src/main/AndroidManifest.xml:18](#): Redundant label can be removed

```
15 <activity
16   android:name=".MainActivity"
17   android:exported="true"
18   android:label="@string/app_name">
19   android:theme="@style/Theme.AppCompat">
20   <intent-filter>
21     <action android:name="android.intent.action.MAIN" />
```

RedundantLabel Correctness Warning Priority 5/10

EXPLAIN DISMISS



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Obsolete Gradle Dependency

[././build.gradle.kts:39:](#) A newer version of androidx.appcompat.appcompat than 1.6.1 is available: 1.7.0

```
36
37 dependencies {
38
39     implementation("androidx.appcompat.appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout.constraintlayout:2.1.4")
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43 }
```

[././build.gradle.kts:40:](#) A newer version of com.google.android.material:material than 1.11.0 is available: 1.12.0

```
37 dependencies {
38
39     implementation("androidx.appcompat.appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout.constraintlayout:2.1.4")
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43     implementation("androidx.navigation.navigation-ui:2.7.6")
44 }
```

[././build.gradle.kts:41:](#) A newer version of androidx.constraintlayout.constraintlayout than 2.1.4 is available: 2.2.1

```
38
39     implementation("androidx.appcompat.appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout.constraintlayout:2.1.4")
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43     implementation("androidx.navigation.navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45 }
```

[././build.gradle.kts:42:](#) A newer version of androidx.navigation.navigation-fragment than 2.7.6 is available: 2.8.9

```
39     implementation("androidx.appcompat.appcompat:1.6.1")
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout.constraintlayout:2.1.4")
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43     implementation("androidx.navigation.navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46 }
```

[././build.gradle.kts:43:](#) A newer version of androidx.navigation.navigation-ui than 2.7.6 is available: 2.8.9

```
40     implementation("com.google.android.material:material:1.11.0")
41     implementation("androidx.constraintlayout.constraintlayout:2.1.4")
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43     implementation("androidx.navigation.navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46     androidTestImplementation("androidx.test.espresso.espresso-core:3.5.1")
47 }
```

[././build.gradle.kts:45:](#) A newer version of androidx.test.ext:junit than 1.1.5 is available: 1.2.1

```
42     implementation("androidx.navigation.navigation-fragment:2.7.6")
43     implementation("androidx.navigation.navigation-ui:2.7.6")
44     testImplementation("junit:junit:4.13.2")
45     androidTestImplementation("androidx.test.ext:junit:1.1.5")
46     androidTestImplementation("androidx.test.espresso.espresso-core:3.5.1")
47 }
```



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Obsolete SDK_INT Version Check

[././src/main/res/values-v23](#): This folder configuration (v23) is unnecessary; minSdkVersion is 24. Merge all the resources in this folder into values.

ObsoleteSdkInt

Performance

Warning

Priority 6/10

EXPLAIN

DISMISS

Unused resources

[././src/main/res/values/colors.xml](#):3: The resource R.color.black appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <resources>
3   <color name="black">#FF000000</color>
4   <color name="white">#FFFFFF</color>
5 </resources>
```

[././src/main/res/values/colors.xml](#):4: The resource R.color.white appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <resources>
3   <color name="black">#FF000000</color>
4   <color name="white">#FFFFFF</color>
5 </resources>
```

[././src/main/res/layout/content_main.xml](#):2: The resource R.layout.content_main appears to be unused

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
3   xmlns:app="http://schemas.android.com/apk/res-auto"
4   android:layout_width="match_parent"
5   android:layout_height="match_parent">
```

[././src/main/res/values/dimens.xml](#):2: The resource R.dimen.fab_margin appears to be unused

```
1 <resources>
2   <dimen name="fab_margin">16dp</dimen>
3 </resources>
```

[././src/main/res/menu/menu_main.xml](#):1: The resource R.menu.menu_main appears to be unused

```
1 <menu xmlns:android="http://schemas.android.com/apk/res/android"
2   xmlns:app="http://schemas.android.com/apk/res-auto"
3   xmlns:tools="http://schemas.android.com/tools"
4   tools:context="com.example.myapplication.MainActivity">
```

Lint Report: 34 warnings

Check performed at Mon Apr 21 11:16:26 IST 2025 by AGP (8.2.1)

EXPLAIN

DISMISS

Missing inputType

[././src/main/res/layout/activity_main.xml](#):11: This text field does not specify an inputType

```
8   tools:context=".MainActivity">
9   <!-- Edit text to enter course name -->
10   <EditText
11     android:id="@+id/editCourseName"
12     android:layout_width="match_parent"
13     android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):20: This text field does not specify an inputType

```
17   android:layout_width="match_parent"
18   android:layout_height="wrap_content">
19   <!-- Edit text to enter course duration -->
20   <EditText
21     android:id="@+id/editCourseDuration"
22     android:layout_width="match_parent"
23     android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):29: This text field does not specify an inputType

```
26   android:layout_width="match_parent"
27   android:layout_height="wrap_content">
28   <!-- Edit text to display course tracks -->
29   <EditText
30     android:id="@+id/editCourseTracks"
31     android:layout_width="match_parent"
32     android:layout_height="wrap_content">
```

[././src/main/res/layout/activity_main.xml](#):38: This text field does not specify an inputType

```
35   android:layout_width="match_parent"
36   android:layout_height="wrap_content">
37   <!-- Edit text for course description -->
38   <EditText
39     android:id="@+id/editCourseDescription"
40     android:layout_width="match_parent"
41     android:layout_height="wrap_content">
```

TextFields

Usability

Warning

Priority 5/10

EXPLAIN

DISMISS



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Use Autofill

[./src/main/res/layout/activity_main.xml:11: Missing autofillHints attribute](#)

```
8      tools:context=".MainActivity">
9
10     <!--Edit text to enter course name-->
11     <EditText
12         android:id="@+id/idEdtCourseName"
13         android:layout_width="match_parent"
14         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:20: Missing autofillHints attribute](#)

```
17         android:minHeight="48dp" />
18
19     <!--edit text to enter course duration-->
20     <EditText
21         android:id="@+id/idEdtCourseDuration"
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:29: Missing autofillHints attribute](#)

```
26         android:minHeight="48dp" />
27
28     <!--edit text to display course tracks-->
29     <EditText
30         android:id="@+id/idEdtCourseTracks"
31         android:layout_width="match_parent"
32         android:layout_height="wrap_content"
```

[./src/main/res/layout/activity_main.xml:38: Missing autofillHints attribute](#)

```
35         android:minHeight="48dp" />
36
37     <!--edit text for course description-->
38     <EditText
39         android:id="@+id/idEdtCourseDescription"
40         android:layout_width="match_parent"
41         android:layout_height="wrap_content"
```

Autofill

Usability

Warning

Priority 3/10

EXPLAIN

DISMISS



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Hardcoded text

./src/main/res/layout/activity_main.xml:16: Hardcoded string "Enter course Name", should use @string resource

```
13     android:layout_width="match_parent"
14     android:layout_height="wrap_content"
15     android:layout_margin="10dp"
16     android:hint="Enter course Name"
17     android:minHeight="48dp" />
18
19     <!--edit text to enter course duration-->
```

./src/main/res/layout/activity_main.xml:25: Hardcoded string "Enter Course Duration", should use @string resource

```
22     android:layout_width="match_parent"
23     android:layout_height="wrap_content"
24     android:layout_margin="10dp"
25     android:hint="Enter Course Duration"
26     android:minHeight="48dp" />
27
28     <!--edit text to display course tracks-->
```

./src/main/res/layout/activity_main.xml:34: Hardcoded string "Enter Course Tracks", should use @string resource

```
31     android:layout_width="match_parent"
32     android:layout_height="wrap_content"
33     android:layout_margin="10dp"
34     android:hint="Enter Course Tracks"
35     android:minHeight="48dp" />
36
37     <!--edit text for course description-->
```

./src/main/res/layout/activity_main.xml:43: Hardcoded string "Enter Course Description", should use @string resource

```
40     android:layout_width="match_parent"
41     android:layout_height="wrap_content"
42     android:layout_margin="10dp"
43     android:hint="Enter Course Description"
44     android:minHeight="48dp" />
45
46     <!--button for adding new course-->
```

./src/main/res/layout/activity_main.xml:52: Hardcoded string "Add Course", should use @string resource

```
49     android:layout_width="match_parent"
50     android:layout_height="wrap_content"
51     android:layout_margin="10dp"
52     android:text="Add Course"
53     android:textAllCaps="false" />
54
55 </LinearLayout>
```

HardcodedText

Internationalization

Warning

Priority 5/10

EXPLAIN

DISMISS



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Included Additional Checks

This card lists all the extra checks run by lint, provided from libraries, build configuration and extra flags. This is included to help you verify whether a particular check is included in analysis when configuring builds. (Note that the list does not include the hundreds of built-in checks into lint, only additional ones.)

[LIST ISSUES](#) [DISMISS](#)

Disabled Checks

One or more issues were not run by lint, either because the check is not enabled by default, or because it was disabled with a command line flag or via one or more `lint.xml` configuration files in the project directories.

[LIST MISSING ISSUES](#) [DISMISS](#)



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Suppressing Warnings and Errors

Lint errors can be suppressed in a variety of ways:

1. With a `@SuppressWarnings` annotation in the Java code
2. With a `tools:ignore` attribute in the XML file
3. With a `//noinspection` comment in the source code
4. With ignore flags specified in the `build.gradle` file, as explained below
5. With a `lint.xml` configuration file in the project
6. With a `lint.xml` configuration file passed to lint via the `--config` flag
7. With the `-ignore` flag passed to lint.

To suppress a lint warning with an annotation, add a `@SuppressWarnings("id")` annotation on the class, method or variable declaration closest to the warning instance you want to disable. The id can be one or more issue id's, such as `"UnusedResources"` or `{"UnusedResources", "UnusedIds"}`, or it can be `"all"` to suppress all lint warnings in the given scope.

To suppress a lint warning with a comment, add a `//noinspection id` comment on the line before the statement with the error.

To suppress a lint warning in an XML file, add a `tools:ignore="id"` attribute on the element containing the error, or one of its surrounding elements. You also need to define the namespace for the tools prefix on the root element in your document, next to the `xmlns:android` declaration:

```
xmlns:tools="http://schemas.android.com/tools"
```

To suppress a lint warning in a `build.gradle` file, add a section like this:

```
android {  
    lintOptions {  
        disable 'TypographyFractions', 'TypographyQuotes'  
    }  
}
```

Here we specify a comma separated list of issue id's after the `disable` command. You can also use `warning` or `error` instead of `disable` to change the severity of issues.

To suppress lint warnings with a configuration XML file, create a file named `lint.xml` and place it at the root directory of the module in which it applies.

The format of the `lint.xml` file is something like the following:

```
<?xml version="1.0" encoding="UTF-8"?>  
<lint>  
    <!-- Ignore everything in the test source set -->  
    <issue id="all">  
        <ignore path="*/test/*" />  
    </issue>  
  
    <!-- Disable this given check in this project -->  
    <issue id="IconMissingDensityFolder" severity="ignore" />  
</lint>
```

Conclusion: Thus we successfully performed analysis of Android apps using android lint.



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EXPERIMENT 7

Aim: Extraction of APK file from android device.

Theory:

Android Debug Bridge (ADB) is a command-line utility included in the Android SDK that serves as a bridge for developers to interface with Android devices from a host computer. It enables both low-level and high-level operations, including device configuration, application debugging, and system maintenance. ADB operates over USB or TCP/IP, and consists of three components: the client (runs on the computer), the daemon (runs on the device), and the server (manages communication between the two). Developers use ADB not just for app deployment and testing, but also for direct access to log output via Logcat, performance profiling, and issuing shell commands in real-time. It's indispensable for Android development and device maintenance, especially when working with rooted or custom ROM environments.

ADB file extraction allows a user to transfer files from the internal storage of an Android device to a connected computer—particularly useful when direct access to the file system is limited or unavailable. The `adb pull` command is commonly used to extract files or directories, even from protected locations if root access is granted. For example, extracting app data from `/data/data/<package_name>` or copying logs and databases can be achieved effortlessly. It is a crucial capability in mobile forensics, security analysis, and disaster recovery, especially when dealing with data corruption, app crashes, or when a device screen is non-functional. Additionally, it supports automation in CI/CD pipelines for mobile apps by programmatically retrieving logs and test artifacts post-deployment.

Output:

1. adb install command

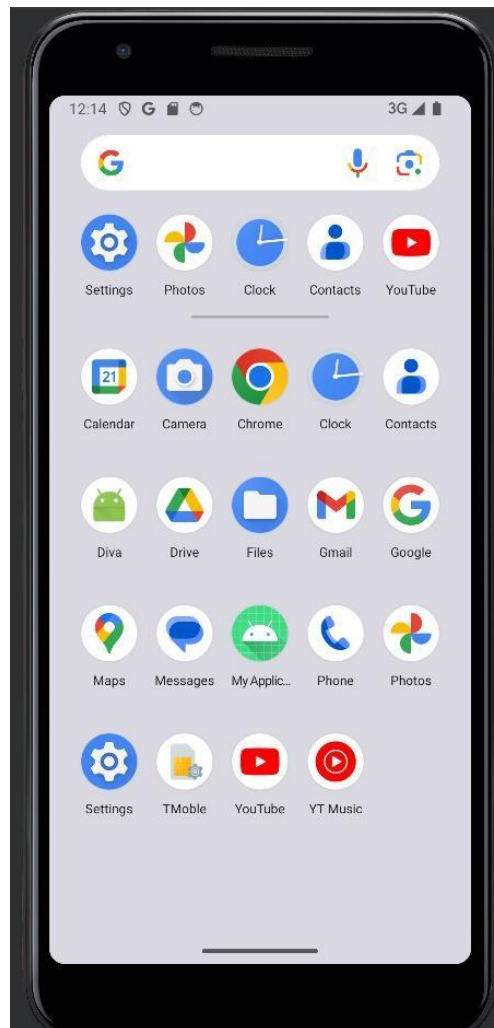


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```
PS C:\Users\djsce.student\AndroidStudioProjects\11> adb install C:\Users\djsce.student\AndroidStudioProjects\11\app\build\outputs\apk\debug\app-debug.apk
Performing Streamed Install
Success
```

2. list packages using adb



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```
PS C:\Users\djsce.student\AndroidStudioProjects\11> adb shell pm list packages f
package:com.android.musicfx
package:com.google.android.connectivity.resources.goldfish.overlay
package:com.android.carrierdefaultapp
package:com.android.emulator.radio.config
package:com.android.backupconfirm
package:com.android.nfc.auto_generated_rro_product__
package:com.google.android.safetycenter.resources
package:com.google.android.wifi.dialog
package:com.android.nfc
package:com.google.android.gsf
package:com.android.location.fused
package:com.google.android.wifi.resources
package:com.google.android.overlay.pixelconfigcommon
package:com.android.carrierconfig.auto_generated_rro_vendor__
package:com.google.android.nearby.halfsheet
package:com.google.android.federatedcompute
package:com.android.theme.font.otoserifsource
package:com.google.android.overlay.largescreenconfig
package:com.android.carrierconfig
package:com.android.systemui.emulation.pixel_fold
package:com.android.carrierconfig.auto_generated_rro_product__
```

3. pull packages using adb

```
PS C:\Users\djsce.student\AndroidStudioProjects\11> adb pull /product/app/YouTube/YouTube.apk
/product/app/YouTube/YouTube.apk: 1 file pulled, 0 skipped. 50.2 MB/s (127097754 bytes in 2.415s)
PS C:\Users\djsce.student\AndroidStudioProjects\11> adb pull /product/app/YouTube/YouTube.apk C:\Users\djsce.student\Documents\Devarsh
/product/app/YouTube/YouTube.apk: 1 file pulled, 0 skipped. 79.6 MB/s (127097754 bytes in 1.523s)
```

4. file pulled successfully

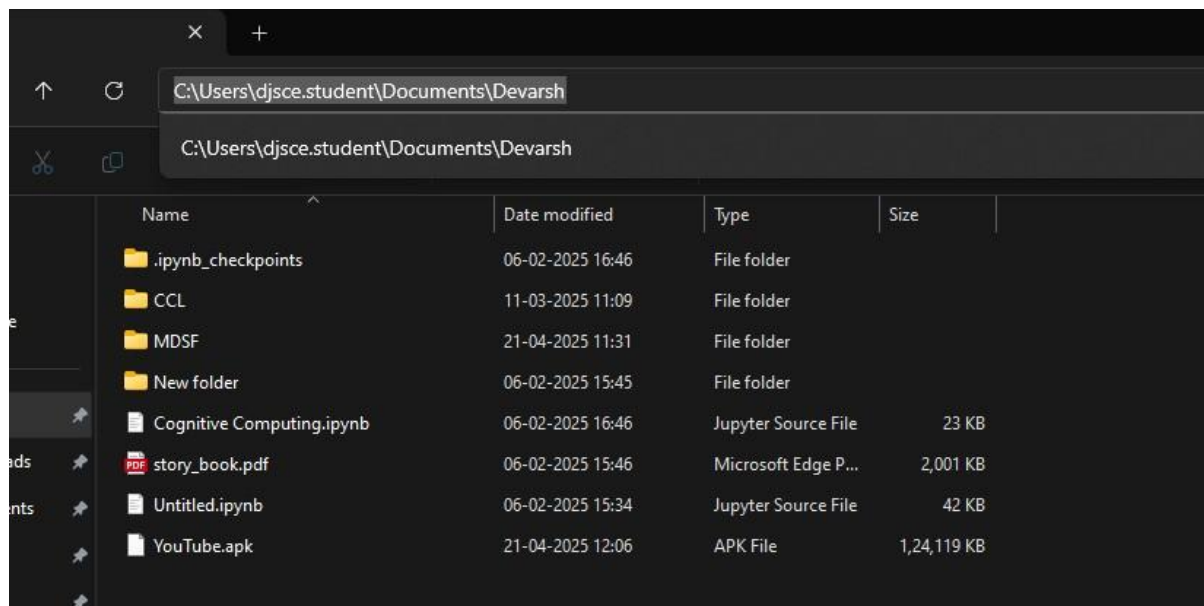


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Conclusion: Thus we successfully Extracted APK file from android device using adb.



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EXPERIMENT 8

Aim: Study of Mobile forensic tools.

1. Cellebrite UFED (Universal Forensic Extraction Device)

- **Overview:**

Cellebrite UFED is a comprehensive mobile forensic solution designed to extract, decode, and analyze data from mobile devices, SIM cards, memory cards, and cloud sources. It supports both logical and physical extractions, and is capable of bypassing device locks on many models.

- **Key Features:**

- Physical and logical extraction from iOS and Android devices.
- Bypass lock screens on certain devices.
- Cloud data acquisition capabilities.
- Data decoding and reporting for court-admissible evidence.

- **Primary Use:**

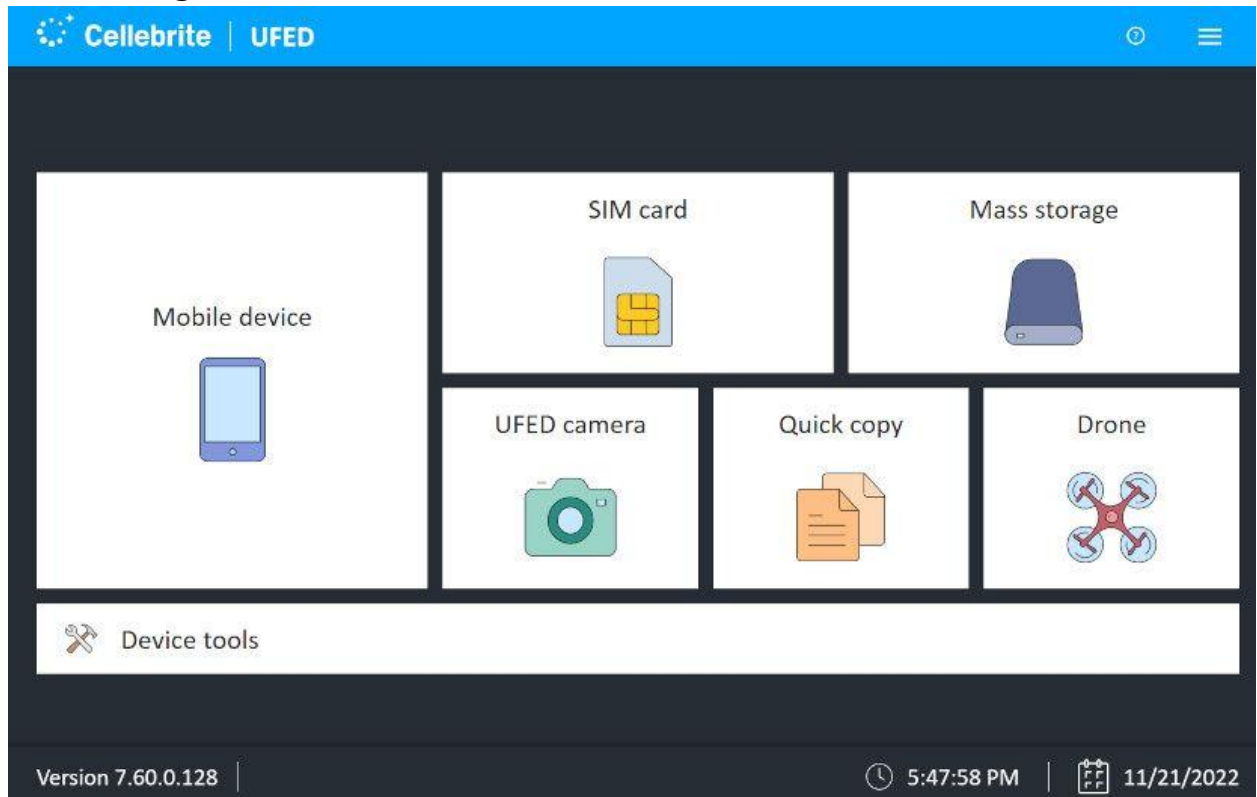
Frequently used by law enforcement, military, and intelligence agencies for criminal investigations and digital evidence collection.



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- **Official Image:**



2. Magnet AXIOM

- **Overview:**

Magnet AXIOM is a digital forensics platform that recovers and analyzes evidence from mobile devices, computers, and cloud services. It organizes recovered data into timelines, connections, and artifacts for investigative purposes.

- **Key Features:**

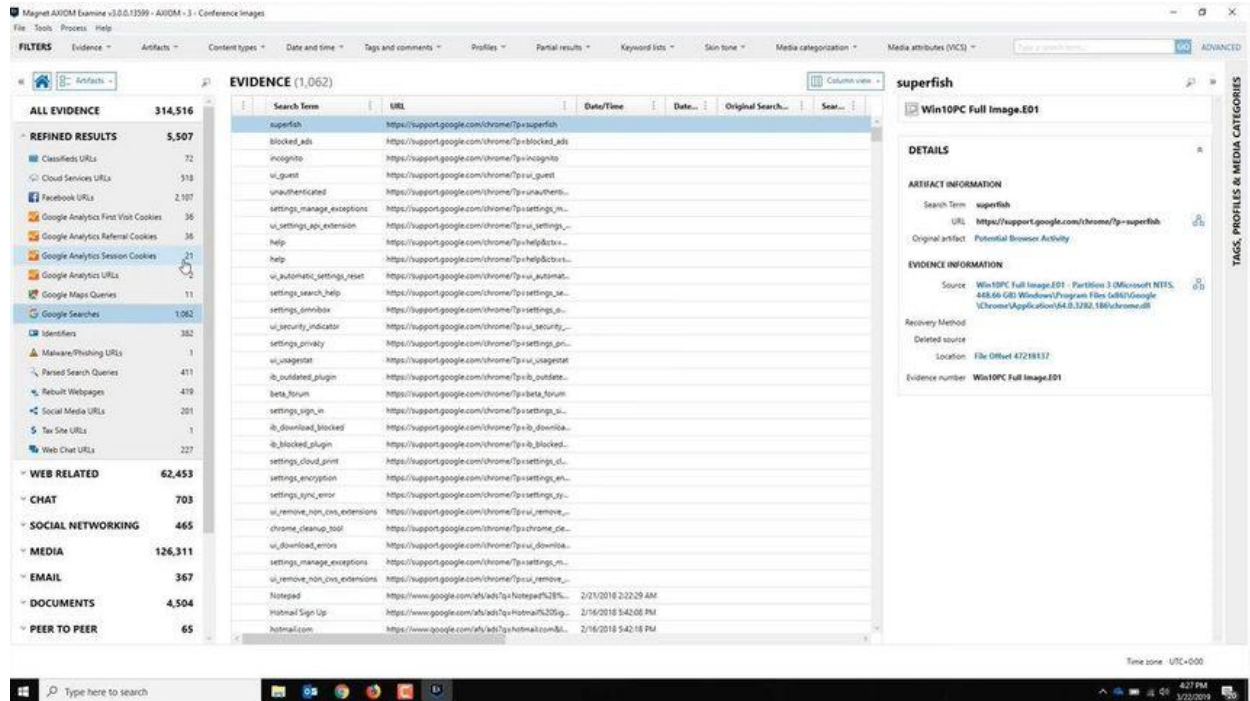
- Comprehensive mobile and cloud data recovery.
- Intuitive visualizations (timeline, connections, artifact explorer).
- Support for encrypted data recovery.
- Court-ready reporting tools.



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- **Primary Use:**
Ideal for investigations requiring correlation of digital evidence across devices and platforms, including criminal cases, corporate investigations, and cybersecurity incidents.
- **Official Image:**



3. Oxygen Forensic Detective

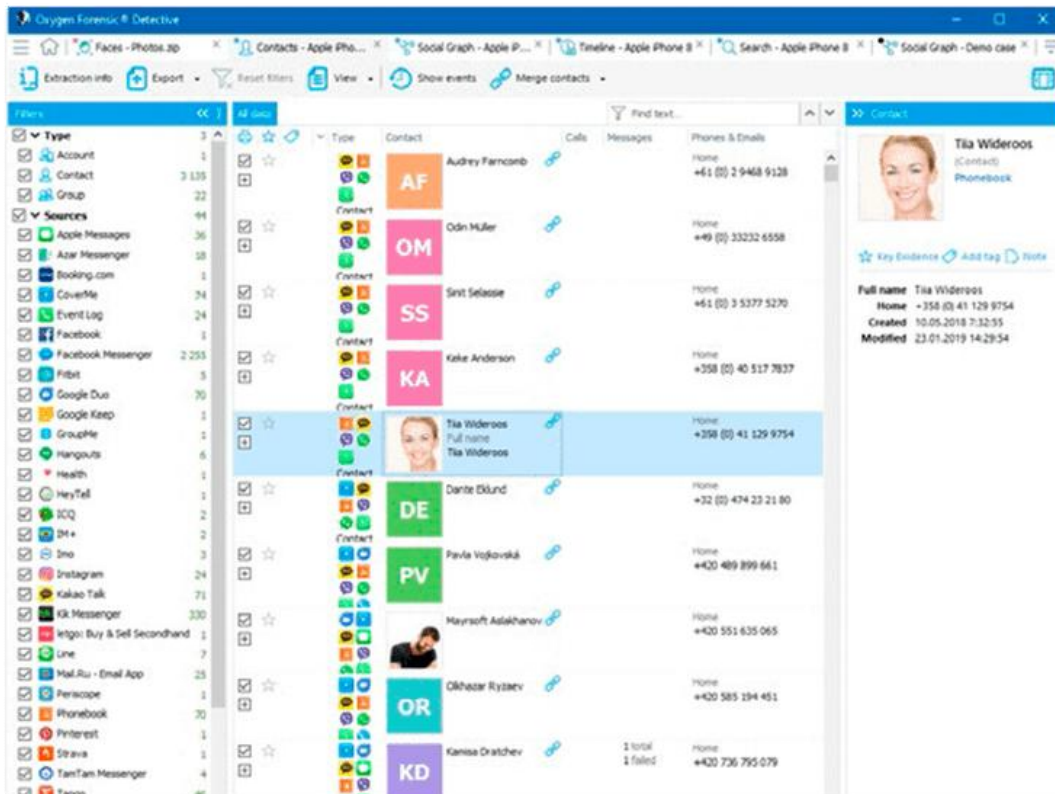
- **Overview:**
Oxygen Forensic Detective is a comprehensive data extraction and analysis tool for mobile devices, cloud services, and IoT devices. It is particularly strong in password extraction, decryption, and recovery of hidden or deleted information.
- **Key Features:**
 - Data extraction from 40,000+ device profiles and apps.
 - Cloud data acquisition and account analysis.
 - Support for encrypted backups and password recovery.
 - Timeline, map, and link graph visualizations.



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- **Primary Use:**
Used for in-depth investigations where access to protected or cloud-stored data is critical, including corporate espionage, cybercrime, and complex criminal cases.
- **Official Image:**



Conclusion:

The study of three prominent mobile forensic tools — Cellebrite UFED, Magnet AXIOM, and Oxygen Forensic Detective — highlights the evolving sophistication and specialization within the field of mobile device forensics. Each tool offers distinct capabilities: Cellebrite excels at physical extraction and lock bypassing, Magnet AXIOM specializes in cross-platform evidence correlation and intuitive visualization, while Oxygen Forensic Detective provides in-depth cloud and IoT data extraction coupled with powerful decryption abilities.